

Marco Vetrano



About Me

Computational Physicist and Data Scientist.

Specialized in developing scalable **Machine Learning** models for insight extraction. I design and implement **AI** solutions to solve complex analytical problems.

Areas of Expertise

Physics • Machine Learning
Network Analysis • Data Analysis

 marco.vetrano01@gmail.com

 Marco Vetrano

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EXPERIENCE

2023–Present

Quantum Machine Learning R&D

UNIVERSITY OF PALERMO • Italy 



- Development of hybrid (Quantum-Classical) algorithms for analyzing and classifying large astrophysical datasets (PNRR STILES Project). Code deployment on Leonardo HPC and IBM Quantum devices.

- **Stack:** Python, Linux, Scikit-Learn, PyTorch.

2025

Quantum Machine Learning R&D

IFISC - UNIVERSITY OF THE BALEARIC ISLANDS • Spain 



- Design and implementation of predictive models for chaotic time series analysis. Code deployment on IFISC-owned HPC cluster.

- **Stack:** Python, Linux, Docker, Scikit-Learn, PyTorch.

EDUCATION

2026

PhD in Physics

• University of Palermo 

2026 - Present

Master in AI Development and Engineering

PROFESSIONAI • 

2023

Master's Degree in Physics

• University of Palermo 

2019

Bachelor's Degree in Physical Sciences

• University of Palermo 

LANGUAGES

Italian	C2	Native Speaker	
English	C1	Writing	Reading
		C1	C1
		Listening	Speaking
		C1	C1

TECH STACK

Python	SQL	C++	Docker	Pytorch
Hugging Face	TensorFlow		XGBoost	
Langchain	Git	FastAPI	Linux	AWS
	Quarto	LaTeX	PySpark	

PUBLICATIONS

- 2025** "State estimation with quantum extreme learning machines beyond the scrambling time.", Springer Nature npj Quantum Information 11.1 (2025): 20.
- 2025** "Exoplanetary atmospheres retrieval via a quantum extreme learning machine.", arXiv preprint arXiv:2509.03617.

CERTIFICATIONS

- 2026** Udemy: Machine Learning and Data Science in Python
- 2023** Duolingo English Test - C1 Certified

PROJECTS

2025	QuantumToolBox UNIVERSITY OF PALERMO · Italy 📍 A Python package developed using numpy, scikit-learn, and plotly for simulating quantum dynamics and implementing classification and time-series prediction models. Available on GitHub in the QuantumStuff repository.	
2025-2026	AstroQELM UNIVERSITY OF PALERMO · Italy 📍 An end-to-end Python package for spectral data analysis. Features include h5/txt parsing, pre-processing/compression via scikit-learn, and processing/classification using qiskit, numpy, and scikit-learn. Available in the AstroQELM repository.	
2026	Instagram-Analyzer · Italy 📍 Backend application developed with FastAPI for automated Instagram scraping. It extracts data from user profiles (reels, comments) using Apify actors, calculating metrics like BPM. Includes Hugging Face models for Speech-to-Text and sentiment analysis. Fully containerized.	
2026	Olist-App · Italy 📍 End-to-end analysis of the Olist e-commerce dataset (Kaggle). Containerized with Docker, it uses SQLAlchemy and PostgreSQL for data management. Features three progressive sentiment analysis models (Scikit-Learn, PyTorch, Hugging Face). Ongoing development includes a Streamlit UI and a custom LLM agent.	

OTHER EXPERIENCE

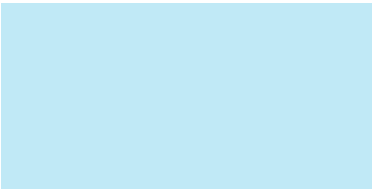
2022 & 2024	Academic Tutor (STEM) UNIVERSITY OF PALERMO · Italy 📍 - Teaching support for Physics, Mathematical Analysis, and Geometry courses. - Personalized mentoring for students with learning disabilities (DSA).
2022	Data Analyst Intern RI.MED FOUNDATION · Italy 📍 Applied Network Analysis techniques to study RNA structures, identifying critical interaction patterns.
2018	Data Analyst Intern IASF - INSTITUTE OF SPACE ASTROPHYSICS AND COSMIC PHYSICS · Italy 📍 Developed C scripts for spectral processing and analysis of black hole data.

AWARDS

2023	First Place - QuHack4AI: Quantum Hackathon for Industrial Applications (University of Bologna & Unipol). Project: Variational algorithms for hydrogen and water molecules.
2023	"Eduardo Gugino" Award for Master's Thesis: "Information Scrambling in Quantum Extreme Learning Machines".

TALKS

2023	"Information Scrambling in Quantum Extreme Learning Machines for State Estimation" at New Trends in Non-equilibrium Statistical Mechanics
2024	"QELM response to scrambling dynamics and applications to atmospheric retrieval" at International School on Non-Equilibrium Phenomena
2025	"Exoplanetary Retrieval with Quantum Extreme Learning Machines" at the National Congress of the Italian Physical Society



Marco Vetrano  marco.vetrano01@gmail.com  Palermo, Italy  +39 3459460312
 <https://marcovetrano01.github.io/Portfolio/>

I authorize the processing of my personal data contained in this CV in accordance with local laws and the GDPR (EU Regulation 2016/679).